An exceptional four-dimensional unitary Hecke Yang–Baxter operator

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Abstract

We construct an explicit 16×16 unitary Yang–Baxter operator in the class

$$\left[e^{i\pi/3}, \frac{1}{2}, 4\right].$$

This realizes the smallest-dimensional previously unresolved class in the exceptional even-dimensional family left open in Lechner's 2026 classification preprint, and that class corresponds to the ordinary-localization question highlighted by Galindo, Hong, and Rowell for the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl sequence. Writing $\mathbb{C}^4 \cong \mathbb{C}^2 \otimes \mathbb{C}^2$, the associated reflection has a five-word real Pauli–Clifford expansion. Four words combine into an involution M, a fifth involution E anti-commutes with M, and the braid relation restricts the reflection circle $\alpha M + \beta E$ to the four points

$$\alpha^2 = \frac{2}{3}, \qquad \beta^2 = \frac{1}{3}.$$

Choosing $(\alpha, \beta) = (\sqrt{2/3}, -1/\sqrt{3})$ gives the displayed operator. The resulting spectral projection has both unnormalized partial traces equal to $2\mathbf{1}_4$, so normalized matrix trace induces the $\eta = 1/2$ Markov trace. Hence the tensor-space representation induces a faithful representation of $H_n(3, 6)$ for every n, providing an ordinary unitary localization of the $\mathcal{C}(\mathfrak{sl}_3, 6)$ Jones–Wenzl sequence. Together with the known dimension-two obstruction and a dimension-three argument, this proves that the minimum local dimension is four. Consequently, the Rowell–Wang localization conjecture, in the form of GHR Conjecture 1.5, is confirmed for the $\mathcal{C}(\mathfrak{sl}_3, 6)$ generator, a case Galindo, Hong, and Rowell singled out as a possible counterexample.

Keywords. Yang–Baxter operator; Hecke algebra; braid group representation; unitary localization; fusion category.

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1 Introduction and main result

Let V be a finite-dimensional complex Hilbert space. A unitary $R \in \mathrm{End}(V \otimes V)$ is a Yang–Baxter operator when

$$(R \otimes \mathbf{1}_V)(\mathbf{1}_V \otimes R)(R \otimes \mathbf{1}_V) = (\mathbf{1}_V \otimes R)(R \otimes \mathbf{1}_V)(\mathbf{1}_V \otimes R). \tag{1}$$

Lechner [4] classified the possible characters of non-involutive unitary R-matrices with two eigenvalues, up to one exceptional family. In his normalization the two eigenvalues are -1 and q, the (-1)-spectral projection P has normalized trace

$$\eta = \frac{\operatorname{rank} P}{(\dim V)^2},$$

and the classes left unresolved by that analysis are

$$\left[e^{i\pi/3}, \frac{1}{2}, 2m\right] \qquad (m \geq 1).$$

The class in base dimension two is empty; base dimension four was therefore the smallest unresolved member of this remaining family.

The ordinary-localization framework and conjecture were introduced by Rowell and Wang [6]. The specific question here was highlighted in the case study of Galindo, Hong, and Rowell [1, §5.6]. They showed that the Jones–Wenzl representations associated with $\mathcal{C}(\mathfrak{sl}_3, 6)$ have a two-dimensional unitary quasi-localization and an 8×8 $(3, 1)$-generalized localization, but no two-dimensional ordinary unitary localization. They wrote that the sequence did not appear to have an ordinary localization and noted it as a possible counterexample to the Rowell–Wang localization conjecture, restated as GHR Conjecture 1.5. Lechner isolated this family as the sole remaining case in his classification preprint and explicitly identified it with the unknown localization arising from the GHR $\mathcal{C}(\mathfrak{sl}_3, 6)$ case study [4, p. 16, discussion following Proposition 3.6]. The present paper supplies an explicit operator in that family. The dimension-four construction and localization proof are independent of Lechner's classification; the minimality argument in Section 6 uses [4, Lemma 3.1 and Theorem 3.4].

The following is the central result of this paper.

Theorem 1.1. *Let*

$$q = e^{i\pi/3} = \frac{1 + i\sqrt{3}}{2}.$$

There is an explicit unitary $R \in \mathrm{End}(\mathbb{C}^4 \otimes \mathbb{C}^4)$ *such that*

$$(R + \mathbf{1}_{16})(R - q\mathbf{1}_{16}) = 0, \qquad R_1 R_2 R_1 = R_2 R_1 R_2,$$

and each eigenvalue $-1, q$ *has multiplicity eight. With the conventions* $\mathrm{Tr}_1 = \mathrm{Tr} \otimes \mathrm{id}$ *and* $\mathrm{Tr}_2 = \mathrm{id} \otimes \mathrm{Tr}$*, its* (-1)*-spectral projection* P *satisfies*

$$\mathrm{Tr}_1(P) = \mathrm{Tr}_2(P) = 2\mathbf{1}_4.$$

Therefore

$$\left[e^{i\pi/3}, \frac{1}{2}, 4\right] \neq \varnothing,$$

and the associated tensor-space representations induce faithful embeddings

$$H_n(3, 6) \hookrightarrow \mathrm{End}\big((\mathbb{C}^4)^{\otimes n}\big) \qquad (n \geq 2).$$

Under the identification in [1, §5.6], these embeddings form an ordinary unitary localization of the $\mathcal{C}(\mathfrak{sl}_3, 6)$ *Jones–Wenzl representation sequence.*

Corollary 1.2. *The minimum local dimension of an ordinary unitary localization of this Jones–Wenzl sequence is four.*

The formula for R is given in Section 2; the matrix identities are proved in Sections 3 and 4. Section 5 proves the full localization claim rather than only the two-site spectral statement. The minimality argument is in Section 6. Section 7 records a generalized Yang–Baxter normal form, Section 8 records verification and provenance, and Section 9 discusses related work, novelty, and limitations.

2 The five-word construction

Put $V = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, and define the real matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Here $J = XZ = -iY$, so $\{I, X, Z, J\}$ is the real Pauli–Clifford basis of $M_2(\mathbb{C})$; every tensor word in this basis is a real signed orthogonal matrix. A word such as $ZIZZ$ denotes $Z \otimes I \otimes Z \otimes Z$ on $V \otimes V$. Define

$$\begin{aligned} H = & -\frac{1}{\sqrt{6}}ZIZZ - \frac{1}{\sqrt{6}}ZIJJ - \frac{1}{\sqrt{6}}JIZJ + \frac{1}{\sqrt{6}}JIJZ \\ & -\frac{1}{\sqrt{3}}XIXX, \end{aligned} \tag{2}$$

and

$$R = \frac{q-1}{2}\mathbf{1}_{16} + \frac{q+1}{2}H = \frac{-1+i\sqrt{3}}{4}\mathbf{1}_{16} + \frac{3+i\sqrt{3}}{4}H. \tag{3}$$

Equations (2)–(3) are an exact specification of all 256 matrix entries in the computational basis. The visible identity in the second qubit is not an ordinary two-site identity amplification of a 4×4 Yang–Baxter operator: the active part uses one qubit of the first four-dimensional site and both qubits of the second. Section 7 identifies it instead as a $(3, 2)$-generalized operator.

The relevant multiplication table is

$$X^2 = Z^2 = I, \qquad J^2 = -I,$$

$$XZ = J = -ZX, \qquad XJ = Z = -JX, \qquad JZ = X = -ZJ.$$

Thus every product of tensor words reduces to a signed word over $\{I, X, Z, J\}$.

3 A circle of reflections

The five terms in (2) have a compact algebraic explanation. Set

$$A = ZIZZ, \quad B = ZIJJ, \quad C = JIZJ, \quad D = JIJZ, \quad E = XIXX$$

and

$$M = \frac{-A - B - C + D}{2}. \tag{4}$$

Lemma 3.1. *The operators A, B, C, D, E are real symmetric involutions. The first four commute, $D = ABC$, and E anticommutes with each of A, B, C, D. Consequently*

$$M^2 = E^2 = \mathbf{1}_{16}, \qquad ME = -EM.$$

For real α, β,

$$H(\alpha, \beta) = \alpha M + \beta E$$

satisfies

$$H(\alpha, \beta)^2 = (\alpha^2 + \beta^2)\mathbf{1}_{16}.$$

Proof. The word multiplication table gives every stated commutation relation and $D = ABC$. One may also prove $M^2 = \mathbf{1}$ without expanding. Simultaneously diagonalize A, B, C, and let their joint eigenvalues be $x, y, z \in \{\pm 1\}$. Since $D = ABC$, the eigenvalue of M is

$$\frac{-x - y - z + xyz}{2} \in \{\pm 1\}.$$

The anticommutation $ME = -EM$ then makes the cross term in $(\alpha M + \beta E)^2$ vanish. □

Remark 3.2 (Graph-phase form). If $x = (-1)^a, y = (-1)^b, z = (-1)^c$, then

$$\frac{-x - y - z + xyz}{2} = -(-1)^{ab+ac+bc}.$$

Thus in the joint stabilizer eigenbasis, M is the negative phase associated with the triangle graph, while E flips all three syndrome bits. This explains the four-plus-one support in (2); no graph-state formalism is needed in the proof.

4 The cubic certificate and Yang–Baxter equation

For any $K \in \mathrm{End}(V \otimes V)$, write

$$K_1 = K \otimes \mathbf{1}_V, \qquad K_2 = \mathbf{1}_V \otimes K.$$

The Yang–Baxter equation for (3) will follow from a cubic identity for H.

Proposition 4.1. *On the unit circle $\alpha^2 + \beta^2 = 1$,*

$$H(\alpha,\beta)_1 H(\alpha,\beta)_2 H(\alpha,\beta)_1 - H(\alpha,\beta)_2 H(\alpha,\beta)_1 H(\alpha,\beta)_2 = \frac{1}{3}\big(H(\alpha,\beta)_1 - H(\alpha,\beta)_2\big) \tag{5}$$

if and only if

$$\beta^2 = \frac{1}{3}, \qquad \alpha^2 = \frac{2}{3}.$$

In particular, the four signed points with these squared coordinates are exactly the solutions on this reflection circle.

Proof. Reduce the left side of (5), minus its right side, in the six-letter word basis. Exactly the following 18 words survive:

coefficient	words
$-\alpha(3\beta^2 - 1)/6$	$IIJIJZ$, $JIZJII$, $ZIJJII$, $ZIZZII$
$+\alpha(3\beta^2 - 1)/6$	$IIJIZJ$, $IIZIJJ$, $IIZIZZ$, $JIJZII$
$-\beta(3\alpha^2 - 3\beta^2 - 1)/3$	$IIXIXX$
$+\beta(3\alpha^2 - 3\beta^2 - 1)/3$	$XIXXII$
$+\alpha(\alpha^2 - 2\beta^2)/2$	$JIJZXX$, $JIZJXX$, $XIJXZJ$, $ZIZZXX$
$-\alpha(\alpha^2 - 2\beta^2)/2$	$XIJXJZ$, $XIZXJJ$, $XIZXZZ$, $ZIJJXX$

This is a finite certificate: each row follows by repeated use of the displayed 2×2 multiplication table. Since $\{I, X, Z, J\}$ is a basis of $M_2(\mathbb{C})$, its sixfold tensor words form a basis of $\mathrm{End}((\mathbb{C}^2)^{\otimes 6})$. Hence the residual vanishes if and only if every displayed coefficient vanishes. On $\alpha^2 + \beta^2 = 1$,

$$\alpha^2 - 2\beta^2 = 1 - 3\beta^2, \qquad 3\alpha^2 - 3\beta^2 - 1 = 2(1 - 3\beta^2).$$

Hence all coefficients vanish when $3\beta^2 = 1$. Conversely, if $\alpha = 0$, the coefficient of $IIXIXX$ is $4\beta/3 \neq 0$. If $\beta = 0$, the coefficient of $IIJIJZ$ is $\alpha/6 \neq 0$. If $\alpha\beta \neq 0$, the first coefficient group forces $3\beta^2 = 1$, and the unit-circle condition then gives $\alpha^2 = 2/3$. The claim follows. ☐

Choosing

$$\alpha = \sqrt{\frac{2}{3}}, \qquad \beta = -\frac{1}{\sqrt{3}}$$

in $H(\alpha, \beta)$ gives exactly (2). Lemma 3.1 and Proposition 4.1 therefore give

$$H^* = H, \qquad H^2 = \mathbf{1}_{16}, \qquad H_1H_2H_1 - H_2H_1H_2 = \frac{1}{3}(H_1 - H_2). \tag{6}$$

Every word in (2) contains a traceless factor, so $\operatorname{Tr} H = 0$. The eigenvalues $+1$ and -1 thus both have multiplicity eight.

Proof of the matrix part of Theorem 1.1. Put

$$a = \frac{q-1}{2}, \qquad b = \frac{q+1}{2},$$

so $R = a\mathbf{1} + bH$. Since $H_i^2 = \mathbf{1}$, direct expansion gives

$$R_1R_2R_1 - R_2R_1R_2 = b\left[a^2(H_1 - H_2) + b^2(H_1H_2H_1 - H_2H_1H_2)\right].$$

But

$$\frac{a}{b} = \frac{q-1}{q+1} = \frac{i}{\sqrt{3}},$$

so $a^2 + b^2/3 = 0$. Equation (6) proves the Yang–Baxter equation.

On the $+1$-eigenspace of H, $R = a + b = q$; on the -1-eigenspace, $R = a - b = -1$. Therefore

$$(R + \mathbf{1})(R - q\mathbf{1}) = 0, \qquad R^*R = \mathbf{1},$$

with both eigenvalues of multiplicity eight. The (-1)-spectral projection is

$$P = \frac{\mathbf{1}_{16} - H}{2},$$

so $P = P^* = P^2$, $\operatorname{rank} P = 8$, and $\eta = 8/16 = 1/2$. Finally, each word in H has a traceless factor on each four-dimensional site. Hence

$$\operatorname{Tr}_1(H) = \operatorname{Tr}_2(H) = 0, \qquad \operatorname{Tr}_1(P) = \operatorname{Tr}_2(P) = 2\mathbf{1}_4.$$

☐

5 The full localization statement

We now verify that the construction localizes the whole Jones–Wenzl tower, not merely its two-site generator. Let $H_n(q)$ be the specialized Hecke algebra of type A_{n-1}, generated by $g_1, \ldots, g_{n-1}$, with the braid and far-commutativity relations and

$$(g_i + \mathbf{1})(g_i - q\mathbf{1}) = 0.$$

Its idempotent generators e_i are related to the braid generators by

$$g_i = q\mathbf{1} - (1+q)e_i.$$

The Yang–Baxter and Hecke relations define a representation

$$\rho_n : H_n(q) \longrightarrow \operatorname{End}(V^{\otimes n}), \qquad \rho_n(g_i) = R_i, \qquad \rho_n(e_i) = P_i.$$

Let

$$\tau_n(T) = 4^{-n}\operatorname{Tr}(T)$$

be normalized matrix trace.

Equip $H_n(q)$ with the conjugate-linear involution

$$e_i^* = e_i, \qquad \text{equivalently} \qquad g_i^* = g_i^{-1}.$$

Since $P_i = P_i^*$ and R_i is unitary, each ρ_n is a unital $*$-representation. If $\iota_n : H_n(q) \to H_{n+1}(q)$ is the standard inclusion, then

$$\rho_{n+1}(\iota_n(x)) = \rho_n(x) \otimes \mathbf{1}_4. \tag{7}$$

For the $\eta = 1/2$ Markov trace, write

$$\operatorname{Ann}_n = \left\{ x \in H_n(q) : \operatorname{tr}_{1/2}^{(n)}(yx) = 0 \text{ for every } y \in H_n(q) \right\}.$$

Proposition 5.1. *The traces $\tau_n \circ \rho_n$ are the Markov traces with parameter $\eta = 1/2$:*

$$\tau_{n+1}((\rho_n(x) \otimes \mathbf{1}_4)P_n) = \frac{1}{2}\,\tau_n(\rho_n(x)) \qquad (x \in H_n(q)). \tag{8}$$

For every n,

$$\ker \rho_n = \operatorname{Ann}_n.$$

Consequently ρ_n induces an injective representation of the semisimple quotient

$$H_n(3,6) = H_n(q)/\operatorname{Ann}_n.$$

Proof. Taking the partial trace over the last tensor factor and using $\operatorname{Tr}_2(P) = 2\mathbf{1}_4$ gives

$$\begin{aligned}\tau_{n+1}((\rho_n(x) \otimes \mathbf{1}_4)P_n) &= 4^{-(n+1)}\operatorname{Tr}(\rho_n(x)\ \operatorname{Tr}_{n+1}(P_n)) \\ &= 4^{-(n+1)}\,2\operatorname{Tr}(\rho_n(x)) = \frac{1}{2}\tau_n(\rho_n(x)).\end{aligned}$$

Moreover,

$$\tau_{n+1}(\rho_n(x) \otimes \mathbf{1}_4) = \tau_n(\rho_n(x)),$$

so the traces are consistent under the standard inclusions. Together with normalization and traciality, this is the defining Markov property. Uniqueness of the Markov trace with prescribed $\operatorname{tr}_{1/2}^{(n)}(e_i) = 1/2$ identifies it with the trace defining $H_n(3,6)$ [7, Eq. (3.2)]. Explicitly, in the normalization used for this quotient,

$$\eta_{6,3} = \frac{1-q^{-2}}{(1+q)(1-q^{-3})} = \frac{1}{2}, \qquad q = e^{i\pi/3}.$$

This is the Wenzl Markov parameter for $H_n(3,6)$. GHR §5.6 identifies this quotient tower with the $\mathcal{C}(\mathfrak{sl}_3,6)$ Jones–Wenzl sequence, and Theorem 5.28 uses that identification for its generalized localization.

The equality $\tau_n \circ \rho_n = \operatorname{tr}^{(n)}_{1/2}$ now gives both kernel inclusions explicitly. If $x \in \mathrm{Ann}_n$, then

$$\begin{aligned} x \in \mathrm{Ann}_n &\Longrightarrow 0 = \operatorname{tr}^{(n)}_{1/2}(x^*x) = \tau_n(\rho_n(x)^*\rho_n(x)) \\ &\Longrightarrow \rho_n(x) = 0, \end{aligned}$$

because normalized matrix trace is faithful. Conversely, for every $y \in H_n(q)$,

$$\rho_n(x) = 0 \Longrightarrow \operatorname{tr}^{(n)}_{1/2}(yx) = \tau_n(\rho_n(y)\rho_n(x)) = 0,$$

so $x \in \mathrm{Ann}_n$. Thus the kernel is exactly the annihilator and the induced map on the quotient is injective. □

Remark 5.2 (Conventions in GHR §5.6). The published text of GHR §5.6 uses two q-normalizations: its opening paragraph prints $q = e^{2\pi i/3}$, whereas the braid eigenvalues and Theorem 5.28 use $q = e^{2\pi i/6}$. We use the latter eigenvalue normalization. The proof of Theorem 5.28 also prints $\eta = (1-q^{-2})/(1+q^3)$, whose denominator vanishes at this value. The Wenzl parameter used in Proposition 5.1 is $\eta_{6,3} = 1/2$.

Equation (7) and Proposition 5.1 give the quotient-tower compatibility explicitly:

$$\begin{aligned} x \in \iota_n^{-1}(\mathrm{Ann}_{n+1}) &\Longleftrightarrow \rho_{n+1}(\iota_n(x)) = 0 \\ &\Longleftrightarrow \rho_n(x) \otimes \mathbf{1}_4 = 0 \\ &\Longleftrightarrow \rho_n(x) = 0 \\ &\Longleftrightarrow x \in \mathrm{Ann}_n\,. \end{aligned}$$

Thus the inclusions descend to the semisimple quotients and give the commuting square

$$\begin{tikzcd} H_n(3,6) \arrow[r, "\bar{\iota}_n"] \arrow[d, "\bar{\rho}_n"'] & H_{n+1}(3,6) \arrow[d, "\bar{\rho}_{n+1}"] \\ \mathrm{End}(V^{\otimes n}) \arrow[r, "T \mapsto T\otimes \mathbf{1}_4"] & \mathrm{End}(V^{\otimes (n+1)}). \end{tikzcd}$$

Galindo, Hong, and Rowell recall the braid-equivariant isomorphism

$$\mathbb{C}\,\rho_X(B_n) \cong \mathrm{End}_{\mathcal{C}}(X^{\otimes n}) \cong H_n(3,6)$$

for the simple tensor generator X of $\mathcal{C}(\mathfrak{sl}_3,6)$, with $\mathrm{FPdim}(X) = 2$ [1, §5.6]. Since the braid image is the entire algebra being localized, the compatible quotient embeddings above are precisely the injective algebra maps required by [1, Definition 1.3]. Equivalently, by [1, Proposition 4.12 and Definition 4.13], they define an ordinary unitary Hilb_f-localization of the Jones–Wenzl sequence.

Corollary 5.3. *The Rowell–Wang localization conjecture [6, Conjecture 3.1, p. 601], as restated in [1, Conjecture 1.5], holds for the simple tensor generator X of $\mathcal{C}(\mathfrak{sl}_3,6)$: one has*

$$\mathrm{FPdim}(X)^2 = 4,$$

and the associated Jones–Wenzl sequence admits an ordinary unitary localization. Hence the case singled out in [1, Remark 6.2(2)] is not a counterexample.

Proof. Theorem 1.1 supplies the asserted localization, and GHR §5.6 gives $\mathrm{FPdim}(X) = 2$. This does not contradict [1, Theorem 5.27]: that result excludes a fiber functor by showing that one would force a localization of dimension exactly $\mathrm{FPdim}(X) = 2$, whereas the present ordinary localization has local dimension four. □

6 Minimality and amplification

Proof of Corollary 1.2. A one-dimensional local space cannot carry the required two-eigenvalue generator. Galindo, Hong, and Rowell prove that no two-dimensional ordinary unitary localization exists [1, Lemma 5.26]. That lemma quotes the two-dimensional classification up to an overall scalar: a representative with spectrum $\{-\chi, \chi q\}$ is normalized by multiplication by χ^{-1}, which leaves its spectral projection and Markov parameter unchanged.

It remains to exclude local dimension three. Suppose such an ordinary unitary localization existed, with local braid operator $F \in \mathrm{End}(\mathbb{C}^3 \otimes \mathbb{C}^3)$. Since a localization intertwines the braid generators,

$$(F + \mathbf{1}_9)(F - q\mathbf{1}_9) = 0.$$

Injectivity at two strands rules out either spectral projection vanishing, so both eigenvalues -1 and q occur. Thus F belongs to one of the dimension-three Hecke classes in [4, Theorem 3.4]. At $q = e^{i\pi/3}$, the possible Markov parameters are $1/3, 1/2, 2/3$. The value $1/2$ is impossible in dimension three because it would require a projection of rank $9/2$. By [4, Lemma 3.1], the normalized matrix character of each such Hecke operator is the corresponding positive Markov trace.

We now exclude the two remaining parameters directly. In $H_3(q)$, put

$$c = \frac{q}{(1+q)^2} = \frac{1}{3}, \qquad T = e_1e_2e_1 - ce_1.$$

The projection-form Hecke relation is

$$e_1e_2e_1 - e_2e_1e_2 = c(e_1 - e_2).$$

For the Markov trace $\mathrm{tr}_\eta^{(3)}$ with $\mathrm{tr}_\eta^{(3)}(e_i) = \eta$, multiply the displayed projection-form Hecke relation on the right by e_2. Cyclicity and the Markov property give

$$\begin{aligned}\mathrm{tr}_\eta^{(3)}(e_1e_2e_1e_2) &= \mathrm{tr}_\eta^{(3)}(e_2e_1e_2) + c\,\mathrm{tr}_\eta^{(3)}(e_1e_2) - c\,\mathrm{tr}_\eta^{(3)}(e_2)\\ &= \eta^2 - c\eta + c\eta^2,\end{aligned}$$

and hence

$$\begin{aligned}\mathrm{tr}_\eta^{(3)}(T^*T) &= \mathrm{tr}_\eta^{(3)}(e_1e_2e_1e_2) - 2c\eta^2 + c^2\eta\\ &= (1-c)\eta(\eta - c).\end{aligned}$$

Thus a parameter-$1/3$ candidate kills T: its normalized matrix trace is faithful on its image, while the displayed squared norm is zero. In the target $\eta = 1/2$ quotient, however,

$$\mathrm{tr}_{1/2}^{(3)}(T^*T) = \frac{1}{18} > 0,$$

so T is nonzero. In particular, the constructed $\eta = 1/2$ Hecke representation does not factor through the Temperley–Lieb quotient.

For the remaining case put $f_i = \mathbf{1} - e_i$. The complemented projections satisfy the same relation with the same c, since

$$f_1f_2f_1 - f_2f_1f_2 = -\big(e_1e_2e_1 - e_2e_1e_2\big) = c(f_1 - f_2),$$

and their Markov parameter is $1 - \eta$. At $\eta = 2/3$, therefore, the candidate kills

$$T^\perp = f_1f_2f_1 - cf_1,$$

whereas in the target quotient

$$\mathrm{tr}^{(3)}_{1/2}\Big((T^\perp)^* T^\perp\Big) = \frac{1}{18} > 0.$$

Neither dimension-three candidate is faithful. Theorem 1.1 supplies dimension four. □

There is also an immediate infinite family. For vector spaces V, W, let $\Sigma_{23} : (V \otimes W)^{\otimes 2} \to V^{\otimes 2} \otimes W^{\otimes 2}$ exchange the middle tensor factors. For $R \in \mathrm{End}(V^{\otimes 2})$ and $S \in \mathrm{End}(W^{\otimes 2})$, define the reshuffled tensor product

$$R \boxtimes S = \Sigma_{23}^{-1}(R \otimes S)\Sigma_{23}.$$

Taking $S = \mathbf{1}_{W^{\otimes 2}}$ with $W = \mathbb{C}^m$ gives a unitary Yang–Baxter operator on the base space $V \otimes W$, of dimension $4m$ [4, §3]. Its two eigenvalues and normalized spectral rank are unchanged. Therefore

$$\left[e^{i\pi/3}, \frac{1}{2}, 4m\right] \neq \varnothing \qquad (m \geq 1). \tag{9}$$

This does not settle the remaining base dimensions $6, 10, 14, \ldots$.

7 A (3,2)-generalized normal form

The visible identity in the second qubit of every word in (2) does have a precise chain-level interpretation, but a coordinate swap is essential. We use the generalized braid placement of [1, Eq. (3.9) and Remark 3.10(2)] and the associated tower and localization conventions of [1, Example 4.23 and Definition 4.24]. Write the i-th four-dimensional site as

$$V_i = \mathbb{C}^2_{a_i} \otimes \mathbb{C}^2_{b_i},$$

let $\Sigma(u \otimes v) = v \otimes u$ be the within-site qubit swap, and put

$$U_n = \Sigma^{\otimes n}.$$

In the factor order obtained after applying U_n,

$$(b_1, a_1, b_2, a_2, \ldots),$$

define

$$K = \frac{q-1}{2}\mathbf{1}_8 + \frac{q+1}{2} K_H$$

and

$$K_H = -\frac{ZZZ + ZJJ + JJZ - JZJ}{\sqrt{6}} - \frac{XXX}{\sqrt{3}}. \tag{10}$$

Let $\rho_R^{(n)}$ denote the standard braid representation generated by the adjacent copies of R.

Proposition 7.1. *The two-site operator factors after the sitewise swap as*

$$U_2 R U_2^* = \mathbf{1}_{b_1} \otimes K_{a_1, b_2, a_2}.$$

The operator $K \in \mathrm{End}((\mathbb{C}^2)^{\otimes 3})$ is unitary and satisfies the $(3,2)$-generalized Yang–Baxter equation

$$(K \otimes \mathbf{1}_4)(\mathbf{1}_4 \otimes K)(K \otimes \mathbf{1}_4) = (\mathbf{1}_4 \otimes K)(K \otimes \mathbf{1}_4)(\mathbf{1}_4 \otimes K) \tag{11}$$

on $(\mathbb{C}^2)^{\otimes 5}$*, together with far commutativity. If* $\rho_K^{(n)}$ *denotes the resulting* $(3,2)$*-generalized braid representation, explicitly*

$$\rho_K^{(n)}(\sigma_i) = \mathbf{1}_2^{\otimes 2(i-1)} \otimes K \otimes \mathbf{1}_2^{\otimes 2(n-i-1)},$$

then for every n *and every* $x \in \mathbb{C}B_n$*,*

$$U_n \rho_R^{(n)}(x) U_n^* = \mathbf{1}_{b_1} \otimes \rho_K^{(n)}(x).$$

Moreover, for the standard inclusion $\iota_n : \mathbb{C}B_n \hookrightarrow \mathbb{C}B_{n+1}$*, the placement formula gives*

$$\rho_K^{(n+1)}(\iota_n(x)) = \rho_K^{(n)}(x) \otimes \mathbf{1}_4.$$

Consequently K *gives a faithful unitary* $(3,2)$*-localization of the tower* $\{H_n(3,6)\}_{n\geq 2}$*.*

Proof. Conjugating the five words in (2) by U_2 puts the identity factor first and leaves exactly (10) on (a_1, b_2, a_2). This proves the factorization and unitarity. On a chain, adjacent copies of K act on triples shifted by two qubits. After factoring out the global spectator b_1, the ordinary Yang–Baxter equation (1) is precisely (11); nonadjacent triples are disjoint, so far commutativity holds. The displayed chain conjugacy follows generator by generator. The same placement formula gives the displayed compatibility with ι_n. Tensoring by a spectator identity does not change the kernel, and Proposition 5.1 proves faithfulness on $H_n(3,6)$. □

Equivalently, any $(3,2)$-generalized operator K on $W^{\otimes 3}$ blocks into an ordinary operator $\mathbf{1}_W \otimes K$ on the local space $W \otimes W$, in the displayed ordering; the ordinary braid relation is $\mathbf{1}_W$ tensored with (11). This elementary blocking observation explains why the present 8×8 active operator yields a 16×16 ordinary one.

There is an exact tensor-structural distinction from the known operator K_{GHR} in [1, Eq. (5.2), Theorem 5.28, and Remark 5.29(1)]. For an operator L on three qubits, define

$$\begin{aligned}\mathcal{Y}_m(L) = {} & (L \otimes \mathbf{1}_{2^m})(\mathbf{1}_{2^m} \otimes L)(L \otimes \mathbf{1}_{2^m}) \\ & - (\mathbf{1}_{2^m} \otimes L)(L \otimes \mathbf{1}_{2^m})(\mathbf{1}_{2^m} \otimes L).\end{aligned}$$

Direct exact calculation gives, with $\|\cdot\|_F$ the Frobenius norm,

	$\|\mathcal{Y}_1(\,\cdot\,)\|_F^2$	$\|\mathcal{Y}_2(\,\cdot\,)\|_F^2$
K_{GHR}	0	48
K	24	0

Thus the displayed GHR operator is of type $(3,1)$, while the present operator is of type $(3,2)$. The distinction concerns tensor placement, not the two matrices as bare normal operators: they have the same two eigenvalues with the same multiplicities and hence are unitarily similar after forgetting tensor structure. The $(3,2)$ shift is what permits the spectator blocking $\mathbf{1}_2 \otimes K$ into an ordinary operator on the local space $\mathbb{C}^2 \otimes \mathbb{C}^2$; the GHR operator does not yield this ordinary blocking in its displayed $(3,1)$ tensor structure.

GHR also records a different $(3,2)$ operator in Example 3.14. Its two eigenvalues have projective ratio i (up to inversion), whereas the ratio for K is $-q = e^{-2\pi i/3}$ (up to inversion). Those two local operators are therefore not unitarily conjugate up to a phase. In particular, being of type $(3,2)$ is not by itself the new feature here; what is new is the exceptional Hecke spectrum together with a faithful localization of the $H_n(3,6)$ Jones–Wenzl tower.

8 Verification and provenance

The public package supplies three supported exact-verification routes: a hardened SymPy route based on the original discovery-era checker; an independently written standard-library route using a separate sparse exact-arithmetic representation and code path over $\mathbb{Q}(\sqrt{2}, \sqrt{3}, i)$; and a matrix-free Pauli-word route checking the complete 18-word certificate in Proposition 4.1. They check the displayed matrix identities, partial traces, both dimension-three obstructions, the six-dimensional three-strand Hecke image, the generalized operator and far commutativity, and the exact comparison with the GHR matrix. All comparisons are exact, and negative tests deliberately mutate each route. The README records the commands and scope. These finite checks do not replace the printed tower-faithfulness proof, Lechner's dimension-three classification step, or the literature-based priority assessment.

The five-word support was originally found by an AI-assisted structured numerical search in a real Pauli–Clifford basis. The discovery code and random seeds were not retained. Accordingly, this paper makes no claim that the search was reproducible, exhaustive, or established uniqueness. The generative-AI systems identified in the declaration below were also used for mathematical exploration and derivation, adversarial proof analysis, verifier development, literature support, and manuscript preparation. Every theorem claim is supported by the printed derivation, and the finite identities are checked by the exact programs. No conclusion depends on accepting model output as evidence or reproducing the original numerical search.

9 Related work, novelty, and limitations

Rowell and Wang introduced the localization framework and conjecture [6]. Galindo, Hong, and Rowell's 2012 advance publication then studied the $\mathcal{C}(\mathfrak{sl}_3, 6)$ sequence, excluded ordinary local dimension two, and supplied quasi- and $(3, 1)$-generalized localizations, but no ordinary dimension-four operator [1, §5.6]. Rowell's quaternionic model and the unitary-braided-vector-space literature provide close prior art [5, 2]. Lechner then identified $[e^{i\pi/3}, 1/2, 2m]$ as the sole remaining existence family up to braid-character equivalence and stated that its dimension-two member is empty [4, p. 16, discussion following Proposition 3.6].

A focused search through 16 August 2026 found no earlier construction realizing $[e^{i\pi/3}, 1/2, 4]$ and no previously published ordinary four-dimensional localization of this Jones–Wenzl sequence. The construction therefore *appears new* and answers the open class identified in [4]. This negative search is not proof of absolute priority; unindexed, unpublished, paywalled, differently parameterized, or concurrent work may exist.

The broad ordinary-localization question dates to Rowell and Wang; the specific $\mathcal{C}(\mathfrak{sl}_3, 6)$ case was highlighted by GHR in 2012, and the exact 16×16 formulation was isolated in 2026. The result does not classify the exceptional family in every even dimension, prove uniqueness, or identify an intertwiner with the known quaternionic $(3, 1)$ model.

Data, code, and source availability

The source, exact verifiers, frozen outputs, and audit materials are maintained in the public source package in the AlecKriebel/Math repository. The curated source package supplies SHA-256 checksums and explicit manuscript and code licenses. The maintained project page links the current edition. No empirical dataset was generated or analyzed. The public source package contains the exact code, frozen outputs, and sources supporting the computational claims.

The version-specific Zenodo record for this manuscript and its exact-verification package is identified by doi:10.5281/zenodo.21971507 [3].

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